

Bhaumik Singh

Full-Stack Software Engineer | Web, Android & Cloud Expert

Email: connect@bhaumicsingh.dev | GitHub: github.com/mic-360 | LinkedIn: linkedin.com/in/bhaumic

SUMMARY

Full-Stack Software Engineer with production experience architecting **scalable, cloud-native systems** across web and Android platforms. Skilled in **end-to-end system design** — from **distributed microservices** and **RESTful/GraphQL APIs** to **CI/CD automation** and **real-time observability**.

SKILLS

Languages	Java, Python, C++, Go, TypeScript, JavaScript, Kotlin, Dart, Bash
Backend	Node.js, Express.js, Flask, Django, ASP.Net, RESTful APIs, GraphQL, tRPC, WebSockets
Frontend	Next.js, React, React Native (Expo), Flutter, Jetpack Compose, Tailwind CSS, Vite
Databases	PostgreSQL, MongoDB, Redis, Firebase, Prisma ORM, Elasticsearch
Cloud	AWS (EC2, S3, Lambda, CloudFront, VPC), GCP/GKE, Docker, Kubernetes, Terraform, Serverless
DevOps	GitHub Actions, Jenkins, Blue-Green / Canary Deployments, Infrastructure as Code, Nginx
Observability	Prometheus, Grafana, ELK Stack, Performance Profiling, Log Aggregation
Architecture	Microservices, Event-Driven Architecture, System Design, Distributed Systems, SOLID, Agile/Scrum

EXPERIENCE

Karkhana, Gurugram

Jan 2025 – Present

SDE I

Full-time

- Architected and launched **Karkhana Hub**, a production-grade makerspace booking platform using **Next.js 16, tRPC, and MongoDB**, serving **2,000+ active users** with end-to-end machine rental and workshop workflows.
- Eliminated frontend-backend contract drift by designing a strict type-safe service layer with **TypeScript and tRPC** in a monorepo, reducing integration bugs by **90%**.
- Engineered a fault-tolerant payment pipeline with **Razorpay** and idempotent state transitions, achieving **100%** reconciliation accuracy and zero duplicate bookings in production.
- Reduced data-fetch latency by **53%** through strategic **Redis** caching on high-read dashboard queries, enabling responsive performance under concurrent load.
- Deployed a Dockerized CI/CD pipeline on **AWS (EC2, Lambda)** with **Prometheus** monitoring, achieving sub-minute incident detection and high-availability uptime.

PROJECTS

Multi-Tenant Machine Booking Platform

Website

Makerapp

- Architected a full-stack, multi-tenant booking platform using **Next.js 16, TypeScript, tRPC, Prisma, and MongoDB**, supporting machine rentals, lab onboarding, and workshop scheduling across Admin, Vendor, and User roles with role-based OAuth (Google/LinkedIn).
- Engineered transactional booking with **Razorpay**, **Redis**-cached dashboard queries, and containerized **AWS (EC2, S3, Lambda)** deployment via CI/CD with **Prometheus** observability — achieving 100% payment reconciliation and 53% latency reduction.

Lightweight Security Guardian for Web Applications

GitHub

Mamoru-WAF

- Engineered a zero-dependency Web Application Firewall in **Go**, defending against OWASP Top 10 threats (SQLi, XSS, Path Traversal) with pattern-matched rule evaluation at wire speed.
- Built an interactive TUI with **Bubble Tea** delivering real-time traffic monitoring, event logging, and atomic live rule-reloading — zero dropped connections during configuration changes.

High-Performance Background Job Processing

GitHub

Hakai

- Developed an ultra-fast, cross-platform background worker in **Rust** and **Bun** for high-throughput task execution and file-system operations at native speed.
- Designed a robust FFI bridge coupling Rust's safe concurrency model with Bun's JavaScript runtime, outperforming standard Node.js task runners in benchmark throughput.

PowerShell TUI for System Optimization

GitHub

Winmole

- Built a graphical Terminal UI in **Go** and **PowerShell** that automates Windows system cleaning, update management, and network diagnostics from a single interface.
- Implemented safe-delete algorithms and privilege-escalation management, enabling unattended execution of administrative scripts without manual intervention.

Gesture Navigation Extension for Chrome

[GitHub](#)

TabiNeko

- Engineered a Manifest V3 Chrome extension in **TypeScript** and **Bun** delivering fluid right-click mouse-gesture navigation with sub-frame latency recognition.
- Built a custom real-time canvas overlay and mathematical gesture-recognition engine handling directional tracking and circular movement detection for intuitive browser control.

Multimodal Bug Context Engine

[GitHub](#)

AI Feedbacks

- Created a full-stack platform and Chrome Extension in **Next.js** that auto-captures DOM state, UI screenshots, and network errors at crash time — eliminating manual reproduction steps.
- Integrated **Google Gemini** multimodal AI to synthesize crash contexts into structured, developer-ready issue prompts with semantic search across historical reports.

Private Serverless Family Photo Gallery

[GitHub](#)

ShashinMori

- Architected a cross-platform photo gallery using **Flutter + Riverpod** frontend and **Fastify + TypeScript** backend with Firebase OAuth, enabling secure uploads from legacy Pixel devices via **tus protocol** resumable uploads and **BullMQ** job queue workers.
- Achieved **0.3%** upload failure rate, **43%** database query reduction through **Redis** memoization, and sub-100ms page loads across web and Android via asynchronous Jimp-based image optimisation and staggered gallery rendering.

Precision A–B Playback Controller

[GitHub](#)

Spotify Looper

- Architected a cross-platform Spotify playback controller in **Flutter + Riverpod** with millisecond-accurate A–B looping and secure **OAuth 2.0** Web API integration, sustaining >99% session reliability via automated token lifecycle management.
- Engineered a low-latency loop pipeline with **just_audio**, optimising seek operations to sub-120ms glitch-free transitions while delivering a responsive Material 3 UI across Android and Web (PWA).

Self-Hosted Deployment Platform on Android

[GitHub](#)

MiniShinobi

- Built a self-hosted deployment platform on a rooted Android device via **Termux**, featuring **GitHub OAuth**, real-time SSE build logs, Cloudflare Tunnel auto-provisioning, and a serialised build queue optimised for ARM hardware constraints.
- Engineered a zero-native-compilation backend with **Node.js**, **sql.js** (WebAssembly SQLite), and file-based sessions to bypass ARM incompatibilities, backed by **Nginx** reverse proxy and **PM2** process supervision for production-grade uptime.

Privacy-First Link Shortener with MCP Server

[Live](#)

Shortlinks

- Built a full-stack link shortener in **Next.js** and **TypeScript** that strips tracking parameters (utm_*, fbclid, gclid, igshid, 30+ more) before storage, performs platform detection across 23 services, and serves 307 redirects backed by **Appwrite**.
- Shipped the same core engine across three interfaces — a website, a **REST API**, and a **Model Context Protocol (MCP)** server for AI agents — with SHA-256-hashed identity rate limiting (2 req/s) that never persists raw IP addresses.

Private Cinema Streaming App for Google Drive

[Live](#)

HikariPlay

- Built a full-stack **Svelte 5 + SvelteKit** private cinema application in a **Turborepo** Bun monorepo — indexes a Google Drive folder recursively (custom `pLimit(6)` concurrency limiter), resolves poster art via a four-stage **AniList GraphQL** → **OMDb** → **Drive** → **placeholder** fallback with dual-layer caching (24h server in-memory Map + 30-day client **IndexedDB**), and byte-proxies video through the server for Range-request-aware streaming.
- Engineered a two-strategy stream resolver (**Drive API** `alt=media` key auth → **public /uc** **confirmation-HTML** fallback), a **Fuse.js** fuzzy-search gallery with **Svelte 5** `$state/$derived` runes, **GSAP 3 ScrollTrigger** landing-page animations, and a zero-server **localStorage** resume tracker — deployed as a **PWA** on **Cloud Run**.

AI Virtual Try-On Plugin for WooCommerce

[GitHub](#)

Tryon Modal

- Built a WooCommerce plugin delivering in-page AI virtual try-on via **Google Gemini** multimodal API, compositing user photos with garment images through a category-aware prompt pipeline and persisting session history in a custom **MySQL** table.
- Engineered a security-hardened, HPOS-compatible **PHP** plugin with CSRF, SQLi prevention, and XSS escaping; paired with a zero-dependency **vanilla JS** modal featuring lazy-loading, drag-and-drop uploads, and in-modal Add to Cart.

EDUCATION

Institute of Technical Education & Research, SOA University

Bhubaneswar, Odisha

B.Tech — Computer Science & Engineering

Relevant Coursework: Data Structures & Algorithms • Operating Systems • Database Management Systems • Data Science • Machine Learning • Software Engineering •

CERTIFICATIONS

Cybersecurity Professional Specialization

Google **Completed** Aug 2024

Automate Cybersecurity Tasks with Python	Google	<i>Completed</i>	Jun 2024
Tools of the Trade: Linux and SQL	Google	<i>Completed</i>	Mar 2024
Connect and Protect: Networks and Network Security	Google	<i>Completed</i>	Jan 2024
Google Analytics Certificate	Uls Association	<i>Completed</i>	Jan 2024
Frontend Certification Exam	Codedamn	<i>Completed</i>	Mar 2023
Android Mobile Application Development	Meta	<i>Completed</i>	Sep 2022
Architecting Google Compute Engine Specialization	Google	<i>Completed</i>	Sep 2022
Elastic Google Cloud Infrastructure: Scaling and Automation	Coursera	<i>Completed</i>	Sep 2022
Reliable Google Cloud Infrastructure: Design and Process	Coursera	<i>Completed</i>	Sep 2022
Essential Google Cloud Infrastructure: Core Services	Coursera	<i>Completed</i>	Sep 2022
GitHub Actions: CI/CD	Codedamn	<i>Completed</i>	Mar 2022